

OTTER T.S. – Final ACRES

CRUISE DESIGN & MARKING INSTRUCTIONS

Sale Number: 21030

Estimated Sale Value: \$120,000.00

Estimated CV Ranges: 14-44%

Maximum Sampling Error: 10%

Based on Cruise Data: Pre-Cruise Plots

Cruise Methods: Sample Tree (STR)

[Reference attached Minimum Tree and Piece Utilization Standards, dated 11/02/2015]

[Total Sale Acres = 557 Acres including 1 acre ROW]

Otter Payment Unit & Acreage Summary Table

P.U. #	Final Acres	Strata	Marking Method	Species
P.U. 01	21	1	CTM	All Species
P.U. 02	34	1	CTM	All Species
P.U. 03	53	1	CTM	All Species
P.U. 04	37	1	CTM	All Species
P.U. 05	26	1	DxSpd*	100% Count
P.U. 06	39	1	CTM	All Species
P.U. 07	16	1	CTM	All Species
P.U. 08	40	1	CTM	All Species
P.U. 09	54	1	CTM	All Species
P.U. 10	80	1	CTM	All Species
P.U. 11	68	1	CTM	All Species
P.U. 12	48	1	CTM	All Species
P.U. 13	27	1	CTM	All Species
P.U. 14	13	1	DxSpd*	100% Count
P.U. 15	(01)	1	CTM (USFS)	All Species
R.O.W	#dummy#			

*** In PU 05 &14, all Oak, Elm, Cedar, Hemlock, White Pine, Red Pine, Paper Birch, Yellow Birch and ANY orange-banded trees will be retained.**

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STRATA 1: (CTM)

Single-Tree Selection Payment Units: 01-04, 06-13 (517Acres).

100% Count Clearcut Payment Units: 05 & 14 (39 Acres).

Payment Unit 15 (R.O.W.) (1 acre) will be marked and cruised by USFS.

Sample Group	Species/Product (Unit)	DBH Range	Sample Interval
HS	Small Hardwood/Pulpwood (02)	5.0"-10.99"	1:500
HL	Large Hardwood/Pulpwood (02)	11.0"+	1:100
SS	Small Sugar Maple/Sawtimber (01)	11.0"-14.99"	1:50
SL	Large Sugar Maple/Sawtimber (01)	15.0"+	1:25
OS	Small Other Hardwood/Sawtimber (01) <i>[includes Red Maple, Basswood, Black Cherry, Yellow Birch, White Birch, & Red Oak]</i>	11.0"-14.99"	1:50
OL	Large Other Hardwood/Sawtimber (01) <i>[includes Red Maple, Basswood, Black Cherry, Yellow Birch, White Birch, & Red Oak]</i>	15.0"+	1:25
FP	Ash/Pulpwood (02) <i>[includes Black Ash & White Ash]</i>	5.0"+	1:50
FW	Ash/Sawtimber (01) <i>[includes Black Ash & White Ash]</i>	11.0"+	1:25
AS	Small Aspen/Pulpwood (02)	5.0"-11.99"	1:100
AL	Large Aspen/Pulpwood (02)	12.0"+	1:50
CP	Mixed Conifer/Pulpwood (02) <i>[includes Balsam Fir, Tamarack, Cedar, Hemlock, White Pine, Scots Pine, White Spruce, Black Spruce, & Norway Spruce]</i>	5.0"+	1:75
PP	Pine/Pulpwood (02) <i>[includes Red Pine & Jack Pine]</i>	5.0"+	1:1
PW	Pine/Sawtimber (01) <i>[includes Red Pine & White Pine]</i>	9.0"+	1:1

Note: Cruise all Aspen, Balsam Fir, Spruce, & Jack Pine as pulpwood.

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Marking Instructions

Hardwood Selection/Sample Tree Cruising (STR): Payment Units: 01-04, 06-13 (517Acres).

ALL merchantable trees to be removed from these Payment Units require two paint marks at DBH on opposite sides of the tree *AND* a corresponding stump mark (see specific Marking Instructions for these Payment Units). Cruise marked trees using the Sample Groups and Sample Intervals above.

A random start should be used for each sample group at the start of marking in each payment unit. Select a random interval that is less than the required sample interval for the first tally in each sample group, thereafter, sample at the specified sample interval.

Maintain an accurate tree count by payment unit for each sample group. Original field records used for tracking tree counts must be with the tally cards or data recorder sheets when each unit is completed. Do not recopy field notes of tree counts or tally cards.

Clearcut/Sample Tree Cruising (STR, 100% Count): Payment Units: 05 & 14 (39 Acres).

ALL merchantable trees within these Payment Units are designated for removal EXCEPT *Oak, Elm, Cedar, Hemlock, Red Pine, & White Pine, Paper Birch, Yellow Birch* (which are designated retain tree species) AND *Leave Trees* (LTs). Merchantable trees (*except Oak, Elm, Cedar, Hemlock, Red Pine, White Pine, Birch (Paper & Yellow)*) to be removed from these Payment Units only require a paint mark at DBH to verify the tree has been counted; a stump mark is *not* required. Designated retain species (*Oak, Elm, Cedar, Hemlock, Red Pine, White Pine, & Birch (Yellow & Paper)*) require a paint mark at DBH AND at the stump *only* when chosen for removal (operability, disease/insect, or other conditions that may be described in the Marking Guides). Any designated retain trees marked with Cut Tree designation will be documented. **Any and all LTs will be marked by USFS.** LTs are to be marked with a single orange band at DBH and a corresponding orange stump mark.

As stated in the “REMARKS” and “MARKING GUIDE” section, no lone Elm or Oak are to be left. LTs will need to be marked around the lone oak or elm, if there is not a designated retain species adjacent to the elm/oak. LTs may also include healthy groups of hardwoods as stated in the Marking Guides. All banded LTs are to be counted and recorded by species and by payment unit during the cruise, preferably with provided Leave Tree Form and turned in with Tally Cards for each payment unit (see specific Marking Instructions for all Payment Units). Cruise marked trees using the Sample Groups and Sample Intervals above.

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Right-of-Way Clearing (USFS mark)/Sample Tree Cruising (STR): Payment Unit 15 (1 acre)

ALL merchantable trees to be removed from this Payment Unit require paint marks at DBH on opposite sides of the tree *AND* a corresponding stump mark. Cruise marked trees using the Sample Groups and Sample Intervals above.

NOTE: Refer to Marking Guides for specific marking instructions for this Payment Unit.

** USFS personnel will be marking Payment Unit 15 - road R.O.W. at a later date.*

NOTE: Government furnished tree marking paint (TMP) will be issued or returned by appointment only, at the WATERSMEET RANGER DISTRICT OFFICE. The primary point of contact for all TMP transactions will be the paint custodian. This person will be identified at the pre-work meeting. BLUE paint has been designated for this sale.

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CRUISE DESIGN & MARKING INSTRUCTIONS

Cruise Design Prepared by: Hayley Jardeen Forestry Technician Date: 02/18/2026
Final Acres Updated by: Hayley Jardeen, Forestry Technician Date: 02/18/2026

Cruise Design Reviewed by: Brandon Pertile, Silviculture Technician Date: 02/18/2026
Cruise Design Reviewed & Updated by: Hayley Jardeen, Forestry Technician Date: 02/18/2026

Marking Contract Narrative by: _____ Date: _____

Cruise Design & Marking Instructions Approved by: Jesse Cox Date: 3/03/2026

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CRUISE DESIGN & MARKING INSTRUCTIONS

MINIMUM TREE AND PIECE UTILIZATION STANDARDS (11/02/2015 revision)

		Minimum Tree Specifications	Minimum Piece Specifications		
SPECIES	PRODUCT	DBH (inches)	Length (feet)	DIB (inches)	Minimum Net Scale in % of Gross
Hardwoods (except aspen)	Sawlogs	11.0	8	9.6-11.5	95% ^{1/}
Hardwoods	Sawlogs	11.0	8	11.6+	50% ^{1/}
Softwood	Sawlogs	9.0	8	7.6	40
All Species	Pulpwood	5.0	8	4.0	70 and reasonably straight ^{2/}

^{1/} In Timber Sale Contracts, all hardwood sawlogs will use 50% minimum net scale in gross %, due to limitations in TIM. Cruising will use 95% minimum net scale for small diameter logs (9.6 – 11.5 DIB).

^{2/} Seventy percent applies to rot, voids, and char. Mechanical type defects such as sweep, crook, spider heart, and ring shake will not be considered.

Reasonably straight - When the true line of a minimum length piece does not deviate more than ½ the inside diameter of the small and plus two inches from a straight line drawn between the centers of the ends of a piece.

The maximum deflection allowed is what will fit in a 20-inch cylinder.

A minimum tree must include at least one piece that meets minimum specifications.

**** Product heights are now measured to the nearest foot. All heights are measured in feet from the ground on the high side of the stump.**

- 1. Conifers – measure total height (to the tip) in feet and the cruise program will calculate the 4” cut-off.**
- 2. Hardwoods (includes Aspen) – measure the height in feet to the 4” top.**
- 3. Sawtimber – cut-off point or “stoppers” will follow our regular practices. Measure sawtimber height in feet. Secondary product height will be measured as described in item 1 or 2 depending on the species.**
- 4. Default dib’s are already built into the equations;**
 pulpwood = 4.0”
 sawtimber = 9.6” (hardwood)
 = 7.6” (mixed conifers, pine)
- 5. On trees with a broken, dead, or deformed top, record the height to the point where the 4” top or the tip would have been if the tree were still growing normally, then record defect to account for the missing volume.**

**OTTAWA NATIONAL FOREST
SILVICULTURAL PRESCRIPTION / MARKING GUIDES**

NEPA DECISION DOCUMENT: Grosbeck VMP	TIMBER SALE: Otter	PU: See below
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STAND INFORMATION												
PU-Ranger District	Comp-Stand	Acres	MA	LTA	ELTP	Habitat Type	Cover Type	LSC	Year of Origin	Size Density	SI	BA
5 – 02	123-1	26	2.2	11	371B	AVO	89	500	1935	6	62	160
	123-4		2.2	11	371B/C	AVO	89	500	1932	9	59	155
	123-9		2.2	11	371B/ 355A	AVO/ATD	89	500	1983	6	74	160
14 – 02	145-4	13	2.1	11	430C	ATD	89	500	1925	9	57	115
	145-5		2.1	11	421/ 423A	TMC	18	840	1925	9	-	165
	145-9		2.1	11	430C	ATD	89	600	1984	6	75	165
	145-2		2.1	11	430C/ 426C	ATD/AVO	89	600	1932	9	52	145

STAND DESCRIPTION
These payment units consist of an aspen overstory continuous throughout. Aspen diameters will range from 6-20" DBH, with the average around 16". Payment unit 5 will have a slightly smaller average diameter for aspen, generally around 10". Sugar/red maple, yellow birch, basswood, and hemlock are present in the overstory as well but generally in limited capacity. The midstory will mostly consist of sugar maple and red maple. Black ash, basswood, and ironwood are also present within the midstory but at much lower stocking than the maple. The understory consists of maple, aspen shoots, ironwood, ash, and scattered balsam fir and spruce. The average basal area for these two payment units is around 150 ft²/ac.

DESIRED CONDITION
An even-aged aspen stand that is growing vigorously and free of competing vegetation. Long lived conifer species will be present but subordinate to the dominant aspen forest type. Accept mixed hardwood species in areas where aspen fails. Maintain small, dense patches of sapling-sized balsam fir, spruce, and hemlock when practicable. Patches should not exceed 20 percent of the unit acreage.

SHORT AND LONG TERM OBJECTIVES
• Apply silvicultural treatments to regenerate and restore a healthy even-aged aspen/birch stands. • Retain remnant long-lived conifer that are healthy and tolerate the changing environment. • Accept that native pathogens occur in retained trees to provide unique habitat niches for cavity nesting birds, insects, and mammals. • Attention is given to maintaining ecological processes and functions in and around riparian areas where they occur in the sale area. • Acceptable stocking would achieve approximately 3200 seedlings per acre for native species.

REMARKS
There are scattered, small, poorly drained pockets, seeps, steep slopes, perched water tables, or shallow bedrock areas; do not designate any trees for removal in these areas. 100% of spruce and balsam volume should be considered due to signs of spruce decline. 15% of hardwood volume should be considered salvage due to overstocked conditions. 60% of aspen volume should be considered salvage due to age. All other contract groups follow collection rate letter dated 1/27/2025.

NEPA DECISION DOCUMENT: Grosbeck VMP	TIMBER SALE: Otter	PU: See table
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MARKING GUIDE:		
Name of Treatment - 4117 (Clearcut w/ Reserves)	Residual BA/Ac – See below	Acres - See table
The resultant stand appearance should have most acres having 0-10% canopy cover. This will ensure adequate sunlight to warm the ground to encourage suckering and provide a long-term free to grow condition for long term establishment of aspen. • Don't cut species such as hemlock, white pine, red pine, cedar, oak, birch (paper & yellow), and elm species unless necessary to facilitate timber harvest operations. The oak and elm need to have adjacent trees retained to maintain the micro site surrounding the targeted reserve tree, this may include high risk aspen.		

DESIGN CRITERIA AND MITIGATION MEASURES
Highlighted Design Criteria, see NEPA for all the criteria • If any stick nests are found (A1) the wildlife biologist should be consulted. • Reserve snags and den trees according to Forest Plan guidelines (p. 2-32); focus on retaining trees with features beneficial to the NLEB such as shaggy bark and cavities. • If any archaeological sites are located during layout or implementation the Archaeologist should be consulted. • Aspen stands with an objective to regenerate aspen with less than 30 ft²/acre of aspen basal area, harvesting will be restricted to the dormant season December 15 through March 15. Within harvest units with an objective to regenerate aspen (including aspen inclusions within non-clearcut units), harvesting operations and site preparation shall maintain small, dense patches of sapling-sized balsam fir, spruce, hemlock, or other conifer species when practicable.

POST HARVEST TREATMENTS		
FACTS Code	Name of Treatment	Prescription
4493	Site Preparation for Naturals	Remove all residual sub-merchantable material except some patches of low to the ground conifer and retain species noted above
4341	Stocking Survey	Survey to ensure adequate regeneration of 3200 TPA on 40% 1/1000 th ac plots. Evaluate diversity planting opportunities
4341	Stocking Survey	Survey to ensure adequate regeneration of 3200 TPA on 40 % 1/1000 th ac plots
4381	Certification for Naturals w S/P	Enter certification once regeneration goal is met

ACTIVITIES SUMMARY		
Year	Facts Activity Code	Description
2033	4117	Clearcut harvest with reserves
2033	4493	Site preparation for natural regeneration
2034	4341	1 st year stocking survey
2036	4341	3 rd year stocking survey
2036	4381	Certification for natural regeneration w/ SP

Prepared by: Eric Lane	Title: Silviculture Forester	Date: 12/17/2025
Reviewed by:	Title:	Date:
Reviewed by:	Title:	Date:
Approved by: Eric Lane	Title: Certified Silviculturist	Date: 12/17/2025

**OTTAWA NATIONAL FOREST
SILVICULTURAL PRESCRIPTION / MARKING GUIDES**

NEPA DECISION DOCUMENT: Grosbeck VMP	TIMBER SALE: Otter	PU: See below
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STAND INFORMATION												
PU-Ranger District	Comp-Stand	Acres	MA	LTA	ELTP	Habitat Type	Cover Type	LSC	Year of Origin	Size Density	SI	BA
1 – 02	123-6	21	2.1	11	420B/ 426C	ATD/AVO	89	500	1932	9	53	165
2 – 02	123-6	34	2.1	11	420B/ 426C	ATD/AVO	89	500	1932	9	53	165
3 – 02	123-6	53	2.1	11	420B/ 426C	ATD/AVO	89	500	1932	9	53	165
	123-4		2.2	11	371B/C	AVO	89	500	1932	9	59	155
4 – 02	123-4	37	2.2	11	371B/C	AVO	89	500	1932	9	59	155
	123-3		2.2	11	351D/ 371C	AVO	89	500	1932	9	73	160
	123-9		2.2	11	371B/ 355A	AVO/ATD	89	500	1983	6	74	160
	123-10		2.2	11	371C/ 355A	AVO/ATD	89	500	1935	6	62	160
6 – 02	123-6	34	2.1	11	420B/ 426C	ATD/AVO	89	500	1932	9	53	165
7 – 02	123-4	16	2.2	11	371B/C	AVO	89	500	1932	9	59	155
	123-6		2.1	11	420B/ 426C	ATD/AVO	89	500	1932	9	53	165
8 – 02	123-3	40	2.2	11	351D/ 371C	AVO	89	500	1932	9	73	160
	123-4		2.2	11	371B/C	AVO	89	500	1932	9	59	155
9 – 02	145-2	54	2.1	11	430C/ 426C	ATD/AVO	89	600	1932	9	52	145
10 – 02	145-2	80	2.1	11	430C/ 426C	ATD/AVO	89	600	1932	9	52	145
11 – 02	145-2	68	2.1	11	430C/ 426C	ATD/AVO	89	600	1932	9	52	145
12 – 02	145-2	48	2.1	11	430C/ 426C	ATD/AVO	89	600	1932	9	52	145
13 – 02	145-3	27	2.1	11	430C/ 426C	ATD/AVO	89	600	1925	6	52	155
	145-4		2.1	11	430C	ATD	89	500	1925	9	57	115
	145-18		2.1	11	430C	ATD	89	600	1925	6	52	155
	145-5		2.1	11	421/ 423A	TMC	18	840	1925	9	-	165
	145-9		2.1	11	430C	ATD	89	600	1984	6	75	165

STAND DESCRIPTION

These northern hardwood stands are sugar maple dominant throughout the overstory. Other overstory species include red maple, basswood, yellow birch, black cherry, red oak, hemlock, aspen, and white/black ash. These species are generally limited in stocking compared to sugar maple but will occur throughout these stands. The midstory is heavily dominated with maple while other species like ash, red oak, and yellow birch occur but in minor amounts. The understory is variable but mostly consists of maple seedlings/saplings. Black cherry, red oak, yellow birch, hemlock, balsam fir, spruce, ash, and elm are also present throughout the understory but generally stocking is limited. Overall quality varies throughout these stands from overstory to understory, some areas of good quality stems and other areas where there's an abundance of poor quality stems. The basal area will vary from 100-200 ft^2/ac , with an average BA of 155 ft^2/ac .

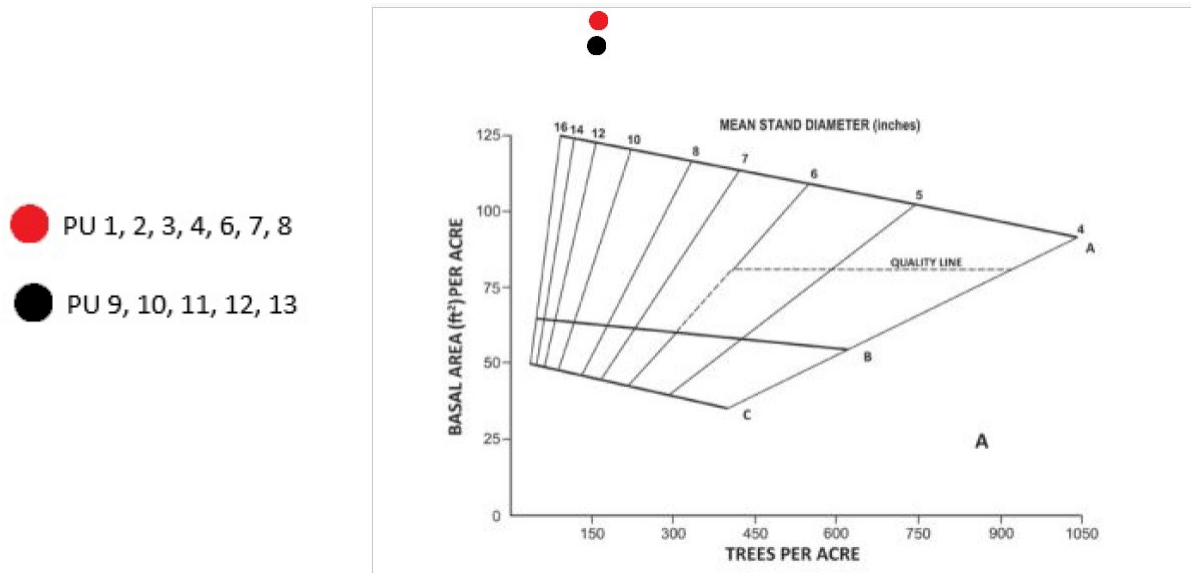


Figure 1. Northern Hardwoods Stocking Chart (Leak et al 2014)

DESIRED CONDITION

Uneven aged northern hardwood stands with the current species diversity maintained or increased. There will be minor amounts of clearcut inclusions, but selection can be applied to areas where the stocking is high and there is no need to regenerate a new age class. In areas where the objective is to regenerate aspen, maintain small, dense patches of sapling-sized balsam fir, spruce, and hemlock when practicable. Patches should not exceed 20 percent of the unit acreage.

SHORT AND LONG TERM OBJECTIVES

- Apply silvicultural treatments to regenerate and restore a healthy uneven-aged northern hardwood stand.
- Retain or increase species diversity by retaining species other than sugar maple.
- Maintain and protect ecological processes and functions in and around riparian areas where they occur within the sale area.
- Promote and maintain aspen inclusions to maintain the health and vigor of the aspen clones.
- Achieve a stocking level of 3,000 seedlings per acre of native species.

REMARKS

There may be scattered, small, poorly drained pockets, seeps, steep slopes, perched water tables, or shallow bedrock areas; do not designate any trees for removal in these areas.

100% of conifer volume should be considered salvage due to spruce decline and I&D issues within the stands. 60% of aspen volume should be considered salvage due to the aspens age class/quality. 15% of hardwood volume should be considered salvage due to overstocked stands and stressed trees. (in reference to collection rate letter dated 1/27/2025).

MARKING GUIDE:		
Name of Treatment - 4151 (Single Tree Selection)	Residual BA/Ac – 90 ft ² /ac	Acres - See table
MARKING GUIDE STANDARDS: • GENERALLY RETAIN: Most hemlock, white pine, red pine, elm, paper birch, oak and cedar unless needed for logging operations. RETAIN ALL ORANGE BANDED TREES. • GENERALLY FAVOR ALL SPECIES OTHER THAN MAPLE AND IRONWOOD. Favor less common species but not at the expense of the best quality trees. If most conditions are relatively equal, continue to improve the relative spacing by marking ironwood, red and sugar maple to meet desired BA. • If advanced regeneration of yellow birch, black cherry, or red oak is present, perform a release to increase sunlight availability (this may include the removal of a large tree, or multiple smaller, poor-quality trees). • Consider all aspen, spruce and balsam high risk and remove all. Where groups of aspen exist, expand opening by one tree length (75') to improve stand diversity. • Look to perform a crown release (ideally 3-4 sided) where healthy red oak occur • Resultant stand should have similar diameter distribution to the original stand with stocking throughout all diameter classes. • Drop zones will be necessary when removing large diameter trees. <u>MARK BY ORDER OF REMOVAL</u> 1. HIGH RISK: Remove trees of any size class that are diseased/structurally damaged that will die within the next 15 years that still contain merchantable products (unless needed for other resource value ie. cavity tree). Identify the best crop trees in the stand and provide a release by marking one or two main competing trees of lesser quality. If BA still higher than desired... 2. DENSITY/SPACING: Improve stand quality and growing space by marking fair – marginal quality trees (mostly 6-10" DBH class). Identify trees that are: • Unacceptable Growing Stock: Trees that are likely to survive for a decade but have structural/growth/disease issues, exhibit signs of marginal improvement or significant volume loss, signs of suppression and are less likely to develop sawlog potential. If BA still higher than desired... Acceptable Growing Stock: Saw log potential trees but possess slight quality issues to form, structure, and soundness (sweep, low branching, old branch stubs, epicormic sprouting). This also includes trees at biological maturity (black cherry > 15", sugar maple and yellow birch > 19", basswood > 21")		

DESIGN CRITERIA AND MITIGATION MEASURES
Highlighted Design Criteria, see NEPA for all of the criteria • Cull trees should be left for wildlife, do not remove trees with little or no value unless they pose a safety hazard. If cull trees need to be marked the silviculturist should be notified to look at the risk. • If any stick nests are found the wildlife biologist should be consulted. • Reserve snags and den trees according to Forest Plan guidelines (p. 2-32); focus on retaining trees with features beneficial to the NLEB such as shaggy bark and cavities. • If any archaeological sites are located during layout or implementation the Archaeologist should be consulted. • Within all harvest units except aspen clearcuts, favor hemlock, white pine, cedar, elm, and oak by retaining and crown releasing these species as directed in the stands' silviculture prescription. These species may be cut to facilitate harvest operations where necessary, or to improve growing space and vigor among these same species that may occur within inclusions. The objective for retaining these species is to improve and/or maintain structural and compositional diversity (Forest Plan, p. 2-2).

POST HARVEST TREATMENTS		
FACTS Code	Name of Treatment	Prescription
4341	Stocking survey	1 st year stocking survey to ensure adequate regeneration of 3000 TPA on 70% of plots – determine tree weed/release timing
4341	Stocking survey	3 rd year stocking survey to ensure adequate regeneration of 3000 TPA on 70% of plots – determine tree weed/release timing
4382	Certification of natural regeneration	Enter certification once regeneration goal is met

ACTIVITIES SUMMARY		
Year	Facts Activity Code	Description
2033	4151	Single tree selection harvest
2034	4341	1 st year stocking survey
2036	4341	3 rd year stocking survey
2036	4382	Certification for natural regeneration

Prepared by: Eric Lane	Title: Silviculture Forester	Date: 12/17/2025
Reviewed by:	Title:	Date:
Reviewed by:	Title:	Date:
Approved by: Eric Lane	Title: Certified Silviculturist	Date: 12/17/2025